

Pulse Duration versus Arm Velocity

Inertial regrasping on the tilted FR3:
what `preview_linear_vel.py` shows, what it leaves out,
and how to choose the velocity for a given gripper pulse

Franka controllers / Genesis regrasp project

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Summary

- The preview releases the bar at the peak of the 0.45 m/s profile and lets it fall at $g \cos \theta$ for 125 ms. In that model 0.45 m/s happens to be almost exactly matched to the pulse at $\theta = 40^\circ$: the bar is caught at the top of its relative motion (26.8 mm of slip, -0.04 m/s relative speed at re-grip).
- That match rests on two assumptions that do not hold. **(1)** The bar is not in free fall. In the tilted pose gravity's transverse part ($g \sin \theta$) presses it onto the lower finger (Genesis kinematics: the transverse direction is the finger axis to within 4°), so the relative motion slows at $g_+ = g(\cos \theta + \mu \sin \theta) \approx 12.2 \text{ m/s}^2$, not 7.5 m/s^2 . **(2)** The arm does not follow the profile. The 0.45 m/s peak falls between two 50 Hz ticks and is never published, and the hand's reversal comes about 28 ms after the commanded one.
- With friction included, the relative motion at 0.45 m/s peaks about 75 ms into the pulse. The rest of a 125 ms pulse gives slip back, and the fingers close on a bar that is sliding down. Genesis shows this directly. At 40° and 0.45 m/s the slip peaks at a 100 ms open window (12.0 mm) and falls to 6.1 mm at 160 ms. At 0.65 m/s it rises to a plateau of 23.6 mm–24.7 mm that no longer depends on pulse length.
- **For a 125 ms pulse the matched peak speed is about 0.65 m/s–0.77 m/s, not 0.45 m/s.** If 0.45 m/s is kept, the matched open time is about 75 ms–85 ms. Both choices give a re-grip at zero relative speed that tolerates ± 10 ms of release timing (Table 2).
- The Robotiq latency still needs measuring. It sets the release instant, and slip changes by roughly 1 mm for every millisecond of early release.

1 What the preview shows

`preview_linear_vel.py` imports `Profile` from `linear_vel_test.py`, so it draws exactly the profile that the test publishes. With the current `DEFAULTS` that profile is:

segment	start [s]	duration [s]	from [m/s]	to [m/s]	rate
ramp up	0.000	0.1125	+0.45	+0.45	4 m/s^2 (from 0)
reverse	0.1125	0.060	+0.45	−0.45	15 m/s^2
reverse cruise	0.1725	0.0867	−0.45	−0.45	hold
stop	0.2592	0.0563	−0.45	0	8 m/s^2

The hand rises 32.1 mm and ends 26.3 mm below its start. The Robotiq pulse is requested at 0.1025 s, which is `gripper_lead_ms` = 10 ms before the reversal begins. The green curve is the preview's model of the released object. It leaves the hand at the reversal with the profile's velocity and then decelerates at $g \cos \theta$ with `tilt_angle` = 40° (7.51 m/s^2) for `FREE_FALL_TIME` = 125 ms.

The preview draws hand and bar in the world frame. The regrasp depends only on their *difference*. The rest of this note works in the hand frame.

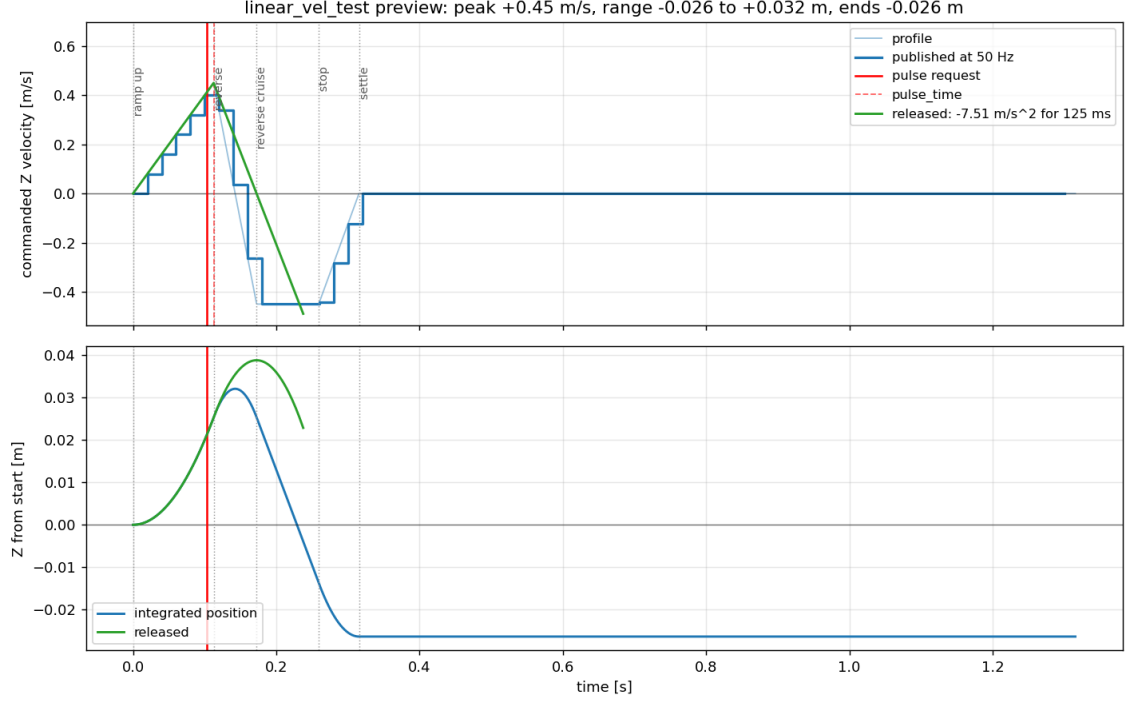


Figure 1: Output of `preview_linear_vel.py` with its defaults, generated by the script itself. Top: commanded velocity, the 50Hz staircase actually published, and the released bar (green). Bottom: integrated positions.

2 Kinematics of a release, seen from the hand

2.1 Definitions

Let s be position along the fixed grasp axis, positive up the axis. The hand moves with $v_h(t)$ and $a_h(t) = \dot{v}_h$. The gripper releases the bar at t_r and re-grips it at $t_r + T_p$, where T_p is the effective open time. Relative quantities are bar minus hand:

$$w(t) = v_o(t) - v_h(t), \quad \Delta(t) = \int_{t_r}^t w \, d\tau, \quad w(t_r) = \Delta(t_r) = 0. \quad (1)$$

$\Delta > 0$ means the bar has moved up the hand, which is the `cuboid_rel_z` that the policy is rewarded for. The quantities that matter are the slip at re-grip, $\Delta(t_r + T_p)$, and the relative speed at re-grip, $w(t_r + T_p)$.

2.2 Free bar (the preview's model)

If only the axial gravity $g_a = g \cos \theta$ acts on the released bar, then

$$\dot{w} = -g_a - a_h(t). \quad (2)$$

Take the preview's case: the release happens at the start of a reversal from $+v_0$ to $-v_1$ at constant deceleration A , taking $T_r = V/A$ with swing $V = v_0 + v_1$, followed by a cruise at $-v_1$. For $T_p \geq T_r$, integrating (2) gives (Appendix A)

$$w(T_p) = V - g_a T_p, \quad (3)$$

$$\Delta(T_p) = V T_p - \frac{V^2}{2A} - \frac{1}{2} g_a T_p^2. \quad (4)$$

The relative motion peaks where $w = 0$. A re-grip at that apex gives the largest slip, zero impact speed, and first-order insensitivity to the pulse length, since $\partial \Delta / \partial T_p = w(T_p)$. The *matched* swing and the slip it produces are

$$V^* = g_a T_p \iff v^* = \frac{1}{2} g_a T_p \quad (v_0 = v_1), \quad \Delta^* = \frac{1}{2} g_a T_p^2 \left(1 - \frac{g_a}{A}\right). \quad (5)$$

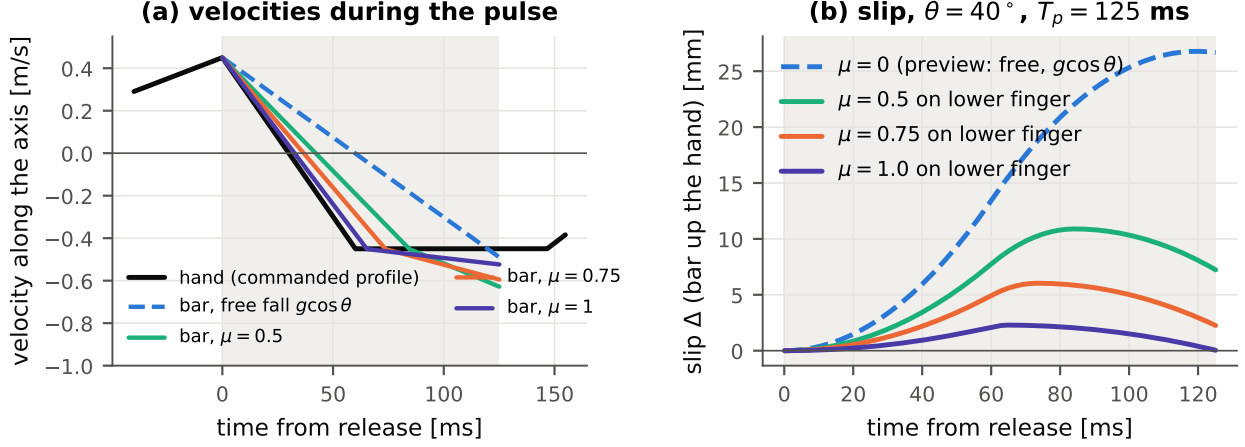


Figure 2: The preview’s profile in the hand frame, using the idealized hand (the commanded profile itself) and a release at the reversal. (a) Velocities: the free-fall bar (dashed) reaches the hand’s cruise speed only at the end of the pulse, while the bar sliding on the lower finger gets there after 65 ms–85 ms. (b) Slip. Without friction the re-grip catches the bar at its apex (26.8 mm). With $\mu = 0.75$ the apex is 6.0 mm at 74 ms, and 2.3 mm is left at re-grip.

Release timing matters much more. Differentiating (4)’s general form with respect to t_r gives

$$\frac{\partial \Delta}{\partial t_r} = w(t_r + T_p) + T_p(a_h(t_r) + g_a). \quad (6)$$

The second term has no counterpart in $\partial \Delta / \partial T_p$. The hand’s acceleration jumps at the reversal, so $\Delta(t_r)$ has a kink there with slopes of opposite sign, and the maximum sits on the kink.

Numbers for the defaults. $V = 0.9$ m/s, $A = 15$ m/s², $T_p = 125$ ms and $g_a = 7.51$ m/s² give $\Delta = 26.8$ mm and $w = -0.039$ m/s. The matched peak speed is $v^* = 0.470$ m/s and $\Delta^* = 29.3$ mm. **In the preview’s own model, 0.45 m/s is matched at 40°.** The timing slopes are +1.40 mm/ms for a release before the reversal (where $a_h = +4$ m/s²) and -0.98 mm/ms for a release after it.

2.3 The bar rests on the lower finger

In the tilted posture the gravity component normal to the axis, $g_n = g \sin \theta$, does not point at empty space. Rotating world gravity into the task frame of `env_franka_parallel_tilted.py` at 40° gives $(-0.49, 6.29, -7.51)$ m/s². The transverse part is anti-parallel to the finger-closing axis (cosine -0.997). So when the fingers open, the bar lies on the lower finger with normal load mg_n and slides against Coulomb friction μmg_n (Figure 2 of the companion RL report). The relative dynamics become piecewise:

$$\dot{w} = \begin{cases} -g_+ - a_h, & w > 0 \text{ (sliding up),} & g_+ = g(\cos \theta + \mu \sin \theta), \\ -g_- - a_h, & w < 0 \text{ (sliding down),} & g_- = g(\cos \theta - \mu \sin \theta), \\ 0, & w = 0 \text{ and } |g_a + a_h| \leq \mu g_n \text{ (stuck).} \end{cases} \quad (7)$$

Three consequences follow for $\theta = 40^\circ$ and $\mu = 0.75$, where $g_+ = 12.24$ m/s² and $g_- = 2.79$ m/s²:

1. **Up-slip needs $A > g_+$.** The bar moves up the hand only while the hand decelerates faster than 12.2 m/s². The commanded 15 m/s² leaves a margin of 2.8 m/s², against 7.5 m/s² in the free model.
2. **The apex comes early and the rest of the pulse is spent sliding back.** Replacing g_a by g_+ in (5) gives $v^* = \frac{1}{2}g_+T_p = 0.765$ m/s for 125 ms. Equivalently, the matched pulse for $v = 0.45$ m/s is $T^* = 2v/g_+ = 73.5$ ms. After the apex the bar slides down at g_- . It sticks only if $\tan \theta \geq 1/\mu$, i.e. $\theta \geq 53^\circ$ for $\mu = 0.75$.
3. **v^* is almost independent of tilt.** $\cos \theta + \mu \sin \theta$ peaks at $\theta = \arctan \mu$, where $v^* = \frac{1}{2}gT_p\sqrt{1+\mu^2}$ (0.77 m/s for $\mu = 0.75$). At 0° it is $\frac{1}{2}gT_p = 0.61$ m/s. For every tilt from 0° to 45° and $\mu \geq 0.5$, a 125 ms pulse needs 0.61 m/s–0.87 m/s (Figure 3).

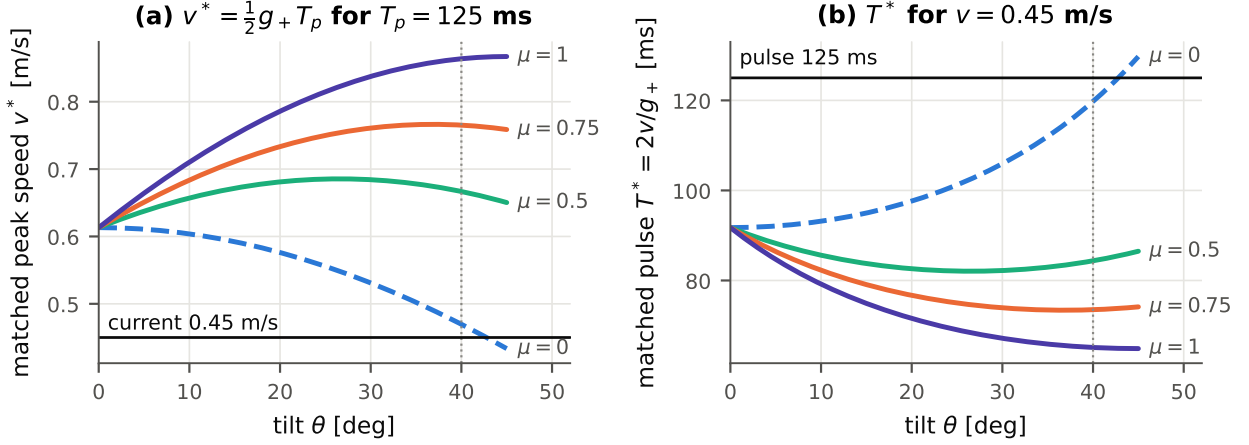


Figure 3: (a) Matched peak speed for a 125 ms pulse versus tilt, from Eq. (5) with g_a replaced by g_+ . The dashed curve is the preview’s frictionless model: it crosses 0.45 m/s only near 40° , the tilt the preview uses. (b) Matched open time for $v = 0.45$ m/s.

3 Validation in Genesis

The model was checked against the simulator the policy was trained in. Two tools were added under `reports/tools/`.

- `pulse_slip_sweep.py` drives `env_franka_parallel_tilted.py` with the same profile shape as `linear_vel_test.py`: ramp at 4 m/s^2 , reverse at 15 m/s^2 , cruise, stop. It fires one pulse per environment. Each environment has its own peak speed, pulse length L (forced-open window $(L - 1) \times 20$ ms) and trigger offset. Finger gain ($k_p = 300$) and friction are pinned, and the pulse delay is set to zero so the forced-open window starts at the trigger.
- `pulse_hires.py` wraps `scene.step` to log hand velocity, bar velocity, in-hand position and finger contact force at 1 kHz, for five friction values. It also measures the angle between the transverse gravity and the finger axis.

Release timing is what friction changes at 0° . At zero tilt the bar is free once released. Yet the slip at a fixed profile went from -4.0 mm ($\mu = 0.25$) through 1.3 mm and 12.2 mm to 21.0 mm ($\mu = 1.0$). The contact force shows why. The residual squeeze carries the bar for a μ -dependent time after the trigger. Extrapolating the bar’s free-flight line back to the hand velocity gives an equivalent release instant of 0, 3, 11 and 24 ms. Feeding those instants and Genesis’ own hand velocity into Eq. (2) reproduces the four slips within 0.3 mm (Figure 4a). This is Eq. (6) observed directly: 24 ms of release delay is worth 25 mm of slip. The optimum lies where the hand actually starts to decelerate, about 25 ms after the commanded reversal.

Friction on the lower finger at 40° and 45° . With the same release instants, the Coulomb model (7) with the simulator’s μ gives 11.4 mm (Genesis 11.3 mm) at 40° and 12.0 mm (Genesis 12.2 mm) at 45° for $\mu = 0.75$. The free-fall model predicts 26.3 mm and 29.3 mm. Across $\mu \in [0.25, 1]$ the Coulomb model stays within 3 mm, while the free model is off by up to a factor of four and has the wrong trend: it predicts more slip for more friction (Figure 4b). Figure 5 of the companion RL report overlays the full 1 kHz trace.

Velocity against pulse length. Figure 5 sweeps peak speed and open window in Genesis, with the trigger at the commanded reversal. At 40° and 0.45 m/s the slip peaks at a 100 ms window (12.0 mm) and falls to 11.3, 8.9 and 6.1 mm at 120, 140 and 160 ms: the pulse outlasts the apex. At 0.65 m/s the slip rises to 23.6, 24.7 and 24.4 mm, a plateau that no longer depends on pulse length. That is the matched condition, reached at the speed Figure 3 predicts. At 0° , with no friction to slow the back-slide, 0.45 m/s collapses from 13.1 mm to -5.1 mm over the same windows.

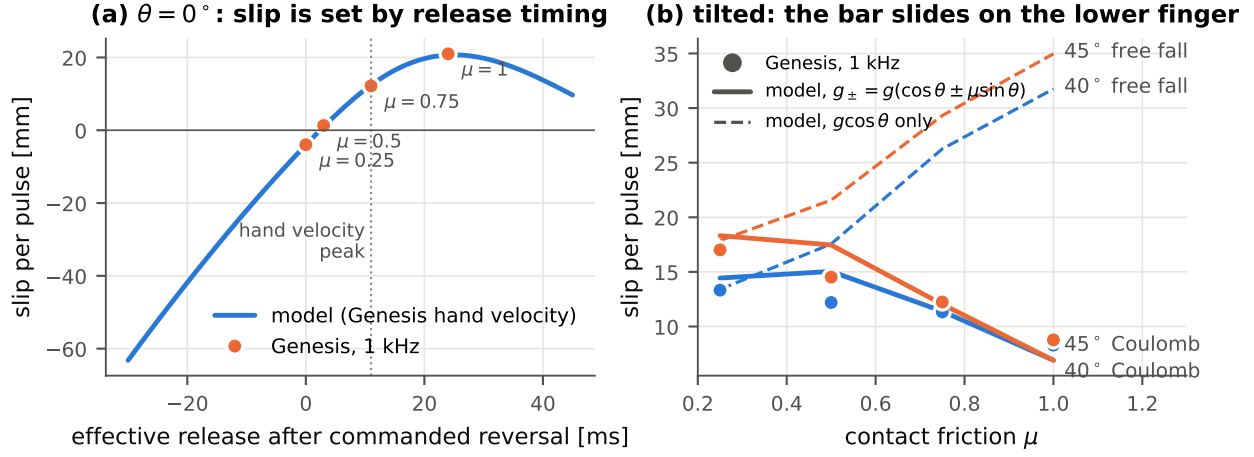


Figure 4: Model versus Genesis at 1 kHz, $v = 0.45$ m/s, 120 ms forced-open window. (a) At 0° the only effect of contact friction is to delay the effective release. The curve is Eq. (2) evaluated with Genesis' hand velocity. (b) At 40° and 45° the free-fall model (dashed) fails. The Coulomb model (7) (solid) matches.

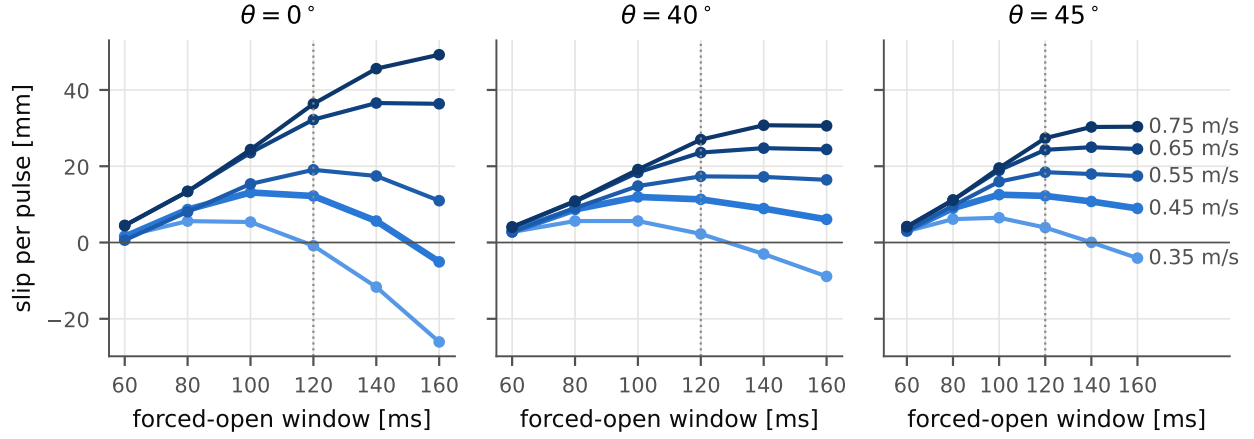


Figure 5: Genesis: slip per pulse against the forced-open window, for peak speeds from 0.35 to 0.75 m/s (the thick line is 0.45 m/s). The dotted line marks the 120 ms window of the current environment. Curves that fall after their peak belong to pulses longer than the apex. Flat tops are matched.

4 What the arm actually does

4.1 The reference chain

The controller does not play the profile. It plays what reaches it, and the default configuration (`config/cartesian_velocity_torque.yaml`) reshapes that in four stages:

1. **Publishing at 50 Hz.** The profile reverses at 112.5 ms, between the ticks at 100 and 120 ms. The published staircase goes $0.40 \rightarrow 0.34$ and **never contains 0.45 m/s**.
2. **interp: cubic, interp_lag: 1.0.** This plays the segment that ends at the newest command, which is one `command_period` (20 ms) late by construction.
3. **Low-pass** ($\omega_n = 400$ rad/s, $\zeta = 0.61$; about 3 ms of delay) and the `max_accel = 19` m/s² slew limit.
4. **The arm's own tracking.** This is modelled with the 2nd-order filter the Genesis environment uses as its stand-in ($\omega_n = 75$ rad/s, $\zeta = 0.33$, tuned against the real arm).

With the defaults, the modelled hand peaks at 0.405 m/s and dips to -0.55 m/s. Its actual reversal comes 28 ms after the commanded one and 38 ms after the pulse request (Figure 6). The preview's green curve starts at 0.45 m/s, 0.045 m/s faster than the hand. Over a 125 ms pulse that difference alone is worth about 5.6 mm of slip.

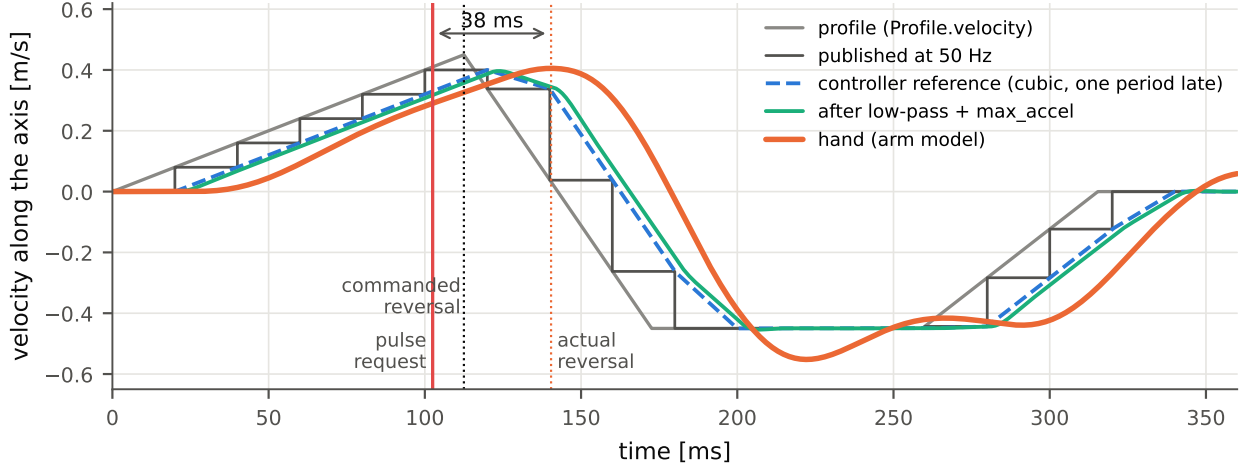


Figure 6: From `Profile.velocity` to the hand, stage by stage. The request goes out 10 ms before the commanded reversal. The hand reverses 38 ms after the request.

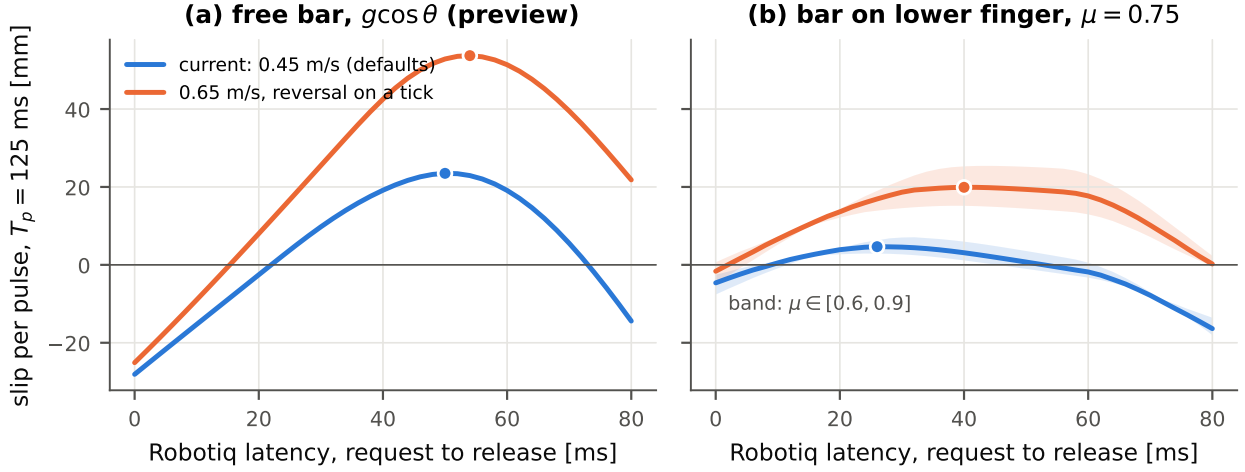


Figure 7: Slip for a 125 ms pulse at 40° against Robotiq latency, with the current 10 ms lead and the modelled hand from Figure 6. (a) Free bar. (b) Bar on the lower finger, with the band covering $\mu \in [0.6, 0.9]$. Dots mark the best latency. At 0.65 m/s the curve in (b) is flat within 1 mm over ± 10 ms of latency.

4.2 Timing budget

The release happens at

$$t_r = t_{\text{rev}}^{\text{cmd}} - t_{\text{lead}} + L_g, \quad (8)$$

where L_g is the Robotiq's latency from service request to the bar coming loose. It has not been measured; the README asks for that measurement before a lead is chosen. Slip is largest when t_r falls on or slightly after the *actual* reversal, so the lead should be

$$t_{\text{lead}} \approx L_g - \tau_{\text{chain}}, \quad \tau_{\text{chain}} \approx 28 \text{ ms}. \quad (9)$$

Figure 7 shows slip as a function of L_g for the current 10 ms lead. With the defaults and $\mu = 0.75$, the best case is only 4.7 mm (at $L_g \approx 26$ ms), and the bar is sliding down at 0.17 m/s when caught. The preview's free-fall model predicts 23.5 mm under the same chain.

5 Why 0.45 m/s is not the right speed for this pulse

The four problems are:

1. **The match in the preview is a coincidence of its assumptions.** Without friction, 0.45 m/s is matched only near 40° (Figure 3a, dashed curve). At 0° the frictionless v^* is already 0.61 m/s.

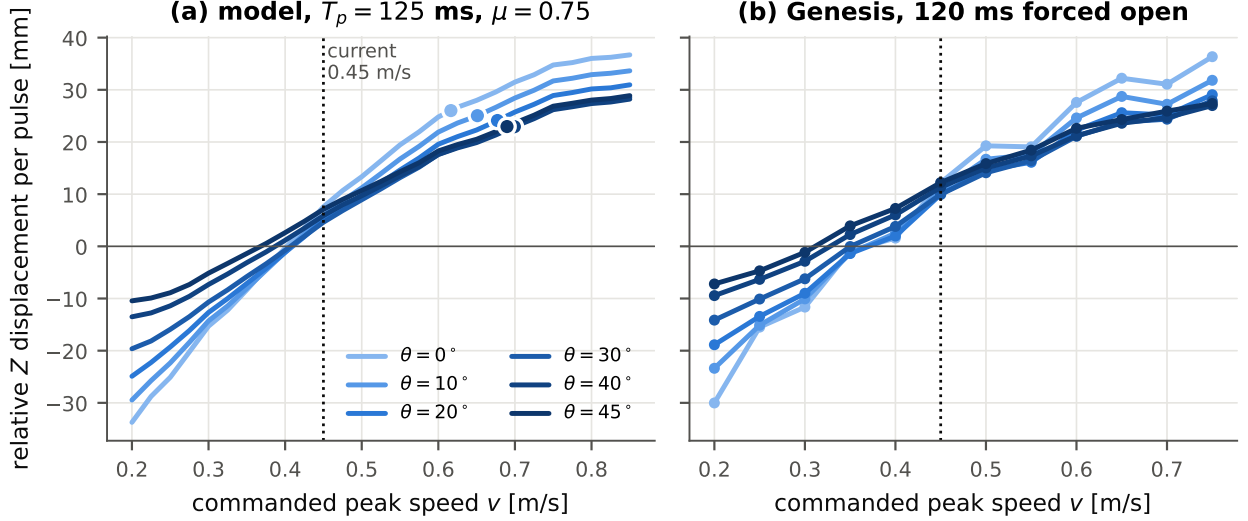


Figure 8: Relative Z displacement (slip up the hand) per pulse against commanded peak speed, for tilts from 0° to 45° . (a) Model through the full controller chain, $T_p = 125$ ms, $\mu = 0.75$, reversal on a publish tick, release on the hand’s actual reversal. Dots mark the matched speed ($w = 0$ at re-grip). (b) Genesis, scripted throws with a 120 ms forced-open window, trigger at the commanded reversal.

2. **With the bar on the lower finger, a 125 ms pulse outlasts the apex by about 50 ms.** The end of the pulse gives slip back, so the fingers close on a bar moving down at 0.15 m/s– 0.2 m/s. The result then depends on the pulse length itself, since $\partial\Delta/\partial T_p = w \approx -0.17$ m/s, or 1.7 mm per 10 ms. Genesis shows the same thing (Figure 5).
3. **The arm never reaches 0.45 m/s.** The reversal falls between publish ticks, and the hand reverses 28 ms late.
4. **A pulse that outlasts the apex is sensitive to everything.** Figure 10 maps slip and “slip lost after the apex” over peak speed and pulse length. The current point sits well inside the region where part of the pulse is wasted. The black line ($w = 0$ at re-grip) crosses $T_p = 125$ ms at about 0.70 m/s (23 mm of slip). At 0.45 m/s the apex is reached with an 85 ms pulse (11.5 mm).

Peak speed across tilts. Figure 8 keeps the current pulse and sweeps the peak speed for $\theta = 0^\circ, 10^\circ, \dots, 45^\circ$. At 0.45 m/s the slip barely depends on tilt: 4.6 mm– 7.6 mm in the model and 9.9 mm– 12.2 mm in Genesis. The reason is that friction and axial gravity trade off: a larger tilt lowers $g \cos \theta$ but raises the friction $\mu g \sin \theta$ on the lower finger. Below about 0.4 m/s the throw loses ground, and the bar ends *lower* in the hand, most of all at small tilts, where nothing holds it during the back-slide. The matched speed, marked with dots, rises from 0.62 m/s at 0° to 0.69 m/s– 0.70 m/s at 30° – 45° , and gives 23 mm– 26 mm of slip. Genesis rises steadily up to the largest speed tested: 36 mm at 0° and 27 mm– 28 mm at 30° – 45° at 0.75 m/s, with no environment resets anywhere in the sweep. In both panels 0.45 m/s sits on the steep, shallow-slip part of every curve.

Friction sweep. The real value of μ between bar and Robotiq pads is unknown, so Figure 9 and Table 1 repeat the matched-speed calculation for $\mu \in [0, 1]$ at each tilt. The closed form is

$$v^*(\theta, \mu) = \frac{1}{2} g T_p (\cos \theta + \mu \sin \theta). \quad (10)$$

It equals the vertical value $\frac{1}{2} g T_p = 0.61$ m/s exactly when $\mu = \tan(\theta/2)$ (0.41 at 45°). Below that friction the tilted grasp needs a slower throw than the vertical one; above it, a faster one. The full-chain values follow the same pattern with a flatter slope, because the hand’s overshoot enlarges the real swing. At $\mu = 0.3$ the matched speed is 0.55 m/s– 0.63 m/s for every tilt. The only corner where 0.45 m/s comes close is a nearly frictionless bar at 40° – 45° ($v^* = 0.46$ – 0.49 m/s at $\mu = 0$). That is the preview’s assumption. The slip at v^* (Figure 9b) falls with friction at large tilts, from about 31 mm at $\mu \leq 0.3$ to 17 mm at $\mu = 1$ for 45° . At 0° it is independent of μ , since there is no normal load.

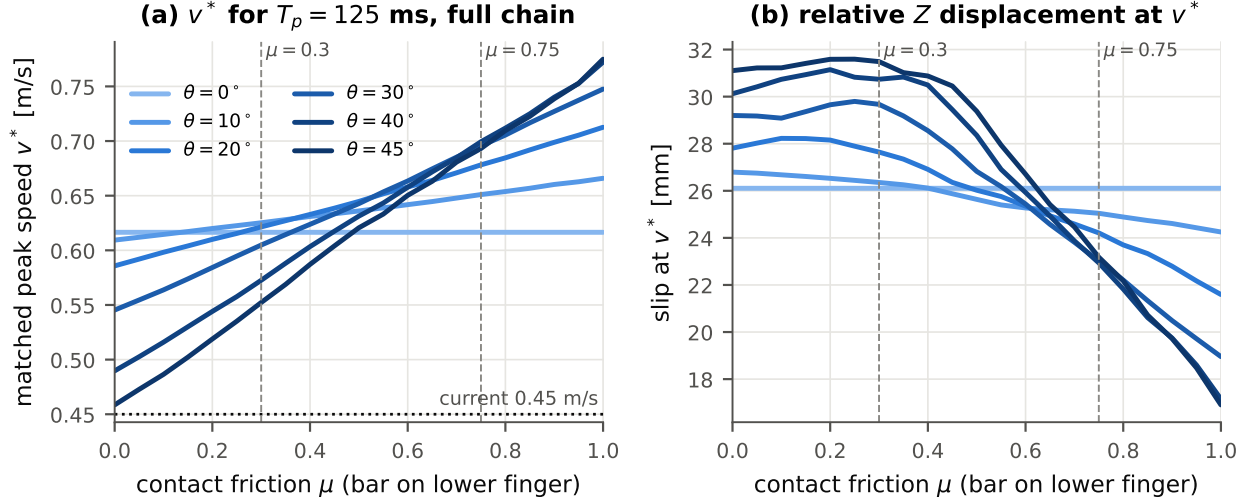


Figure 9: Matched peak speed (a) and the slip it produces (b) against the friction between the bar and the lower finger, one curve per tilt. Full controller chain, $T_p = 125$ ms, reversal on a publish tick, release on the hand's actual reversal. At 0° friction has no effect.

Table 1: Matched peak speed v^* [m/s] for $T_p = 125$ ms: closed form $\frac{1}{2}gT_p(\cos\theta + \mu\sin\theta)$ / full controller chain.

tilt	$\mu = 0$	$\mu = 0.3$	$\mu = 0.5$	$\mu = 0.75$	$\mu = 1.0$
0°	0.61 / 0.62	0.61 / 0.62	0.61 / 0.62	0.61 / 0.62	0.61 / 0.62
10°	0.60 / 0.61	0.64 / 0.63	0.66 / 0.64	0.68 / 0.65	0.71 / 0.67
20°	0.58 / 0.59	0.64 / 0.62	0.68 / 0.64	0.73 / 0.68	0.79 / 0.71
30°	0.53 / 0.55	0.62 / 0.60	0.68 / 0.64	0.76 / 0.70	0.84 / 0.75
40°	0.47 / 0.49	0.59 / 0.57	0.67 / 0.63	0.77 / 0.70	0.86 / 0.77
45°	0.43 / 0.46	0.56 / 0.55	0.65 / 0.62	0.76 / 0.69	0.87 / 0.77

6 Recommendations

1. **Match the speed to the pulse, not the other way round, when the pulse is fixed by the gripper.** For $T_p \approx 125$ ms use about 0.65 m/s, for example

`target_vel:=0.65 accel_up:=3.6111 reverse_hold:=0.15`

which puts the reversal on the 180 ms tick. The model gives 19.9 mm per pulse at zero relative speed, holding within 1 mm over ± 10 ms of latency. Genesis gives 23.6 mm–24.7 mm, flat in pulse length. The costs are more travel (+73/−68 mm, within `max_travel` = 0.3 m) and hand speeds up to 0.71 m/s, below `max_velocity`. The trained policy is capped at 0.45 m/s (`Z_VEL_MAX`), so for closed-loop use it would need retraining with a higher cap.

2. **Or keep 0.45 m/s and shorten the effective open time to about 85 ms.** The model gives 11.6 mm per pulse at zero relative speed, holding within 1.4 mm over ± 10 ms. This suits fine steps; the target range is 12.5 to 22.5 mm.
3. **Put the reversal on a publish tick.** Choose v/a_{up} as a multiple of 20 ms. With the defaults only the tick alignment changes, and the hand peak rises from 0.405 to 0.44 m/s (slip 4.7 to 7.4 mm).
4. **Measure the gripper.** Record the request, the width feedback and the arm twist with `plot_velocity_log.py` for ten pulses. That gives L_g and its jitter, plus the effective open time T_p . Then set $t_{lead} = L_g - 28$ ms.
5. **Measure μ with the gripper open.** Tilt the open hand with the bar resting on the lower finger until it stops sliding: $\mu = 1/\tan\theta_{stick}$. Alternatively, fit μ from the slip of a few pulses using Eq. (7), as done for Genesis in Section 3.
6. **Extend the preview.** Two additions would make its green curve predictive: the $\mu g \sin\theta$ term, and the chain of Figure 6 applied to the hand. Both are about twenty lines, using `controller_chain` and `simulate_slip` in `reports/tools/slip_model.py`.

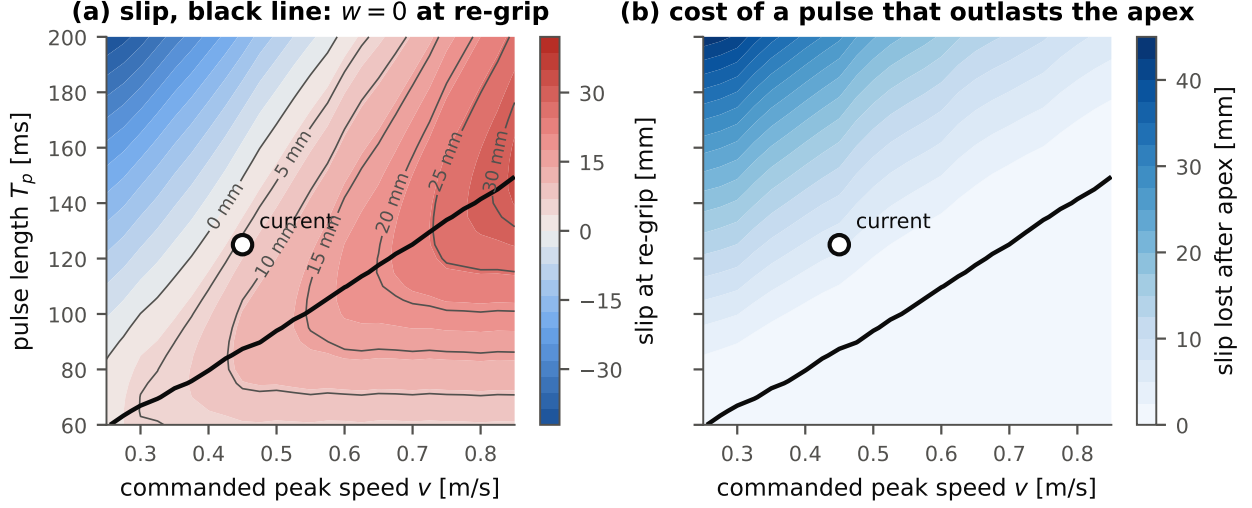


Figure 10: Design map at 40° and $\mu = 0.75$, with the modelled hand and the release on its actual reversal. (a) Slip at re-grip. (b) Slip lost between the apex and the re-grip. The black line is the matched condition $w(T_p) = 0$. Points below it re-grip before the apex, points above it after.

Table 2: Four profiles through the full chain, $\theta = 40^\circ$, $\mu = 0.75$, with the lead set to each profile’s best latency. “Tick” means `accel_up` chosen so that the reversal falls on a publish tick (v/a_{up} a multiple of 20 ms). Columns: slip at best latency; slip at ± 10 ms of latency error; slip for $\mu \in [0.6, 0.9]$; relative speed at re-grip; free-fall slip at its own best latency (the preview’s model); modelled hand peak/trough; commanded travel above/below the start.

profile	v [m/s]	T_p [ms]	L_g^{best} [ms]	slip [mm]	± 10 ms [mm]	μ band [mm]	w [m/s]	free [mm]	hand [m/s]	travel [mm]
current defaults	0.45	125	26	4.7	2.7 / 3.9	2.9–6.3	-0.17	23.5	0.41 / -0.55	+32 / -26
reversal on a tick	0.45	125	33	7.4	5.9 / 5.9	4.8–9.7	-0.21	29.0	0.44 / -0.58	+34 / -32
matched speed	0.65	125	41	19.9	18.9 / 19.3	15.1–25.4	-0.04	53.7	0.64 / -0.71	+73 / -68
matched pulse	0.45	85	45	11.6	10.2 / 11.0	9.2–14.1	-0.01	27.6	0.44 / -0.58	+34 / -32

Limitations. The model is one-dimensional. It ignores bar rotation, finger compliance and the opening and closing transients (Genesis shows these as the μ -dependent release delay). The arm stage borrows the Genesis stand-in instead of a measured real-arm response. The Genesis results use the Franka hand model, not the Robotiq 2F. All of this points the same way: the numbers are best read as trends and orders of magnitude, to be calibrated with the two measurements above.

A Derivation of Eqs. (3)–(6)

Release at $\tau = 0$ at hand speed v_0 . The bar moves as $v_o(\tau) = v_0 - g_a\tau$. The hand decelerates at A until $T_r = V/A$ and then cruises at $-v_1 = v_0 - V$. For $\tau \geq T_r$:

$$\begin{aligned}
 \int_0^{T_p} v_o d\tau &= v_0 T_p - \frac{1}{2} g_a T_p^2, \\
 \int_0^{T_p} v_h d\tau &= v_0 T_r - \frac{1}{2} A T_r^2 - v_1 (T_p - T_r) = \frac{1}{2} (v_0 - v_1) T_r - v_1 (T_p - T_r), \\
 \Delta &= \int_0^{T_p} (v_o - v_h) d\tau = V T_p - \frac{1}{2} V T_r - \frac{1}{2} g_a T_p^2 = V T_p - \frac{V^2}{2A} - \frac{1}{2} g_a T_p^2,
 \end{aligned}$$

and $w(T_p) = v_0 - g_a T_p - (-v_1) = V - g_a T_p$. Setting $w = 0$ gives $V^* = g_a T_p$, and substituting it back gives $\Delta^* = g_a T_p^2 - \frac{g_a^2 T_p^2}{2A} - \frac{1}{2} g_a T_p^2 = \frac{1}{2} g_a T_p^2 (1 - g_a/A)$. For a general hand profile, $\Delta(t_r) = \int_{t_r}^{t_r+T_p} [v_h(t_r) - g_a(t - t_r) - v_h(t)] dt$. The Leibniz rule gives the integrand at the upper limit, which is $w(t_r + T_p)$, minus the integrand at the lower limit, which is 0, plus $\int \partial_{t_r}[\cdot] = T_p(a_h(t_r) + g_a)$. That is Eq. (6). With friction, g_a is replaced by g_+ for as long as $w > 0$.

B Reproduction

```
# Genesis experiments (in the genesis container, workspace at /workspace)
python3 reports/tools/pulse_slip_sweep.py --tilts 0 15 30 45 \
    --out reports/data/pulse_slip.npz
python3 reports/tools/pulse_slip_sweep.py --tilts 0 40 45 \
    --vpeaks 0.35 0.45 0.55 0.65 0.75 --lengths 4 5 6 7 8 9 \
    --offsets -1 0 1 --zvel-max 0.8 --out reports/data/pulse_slip_fast.npz
python3 reports/tools/pulse_hires.py --tilts 0 40 45 \
    --out reports/data/pulse_hires.npz
# figures, table and numbers (host, no Genesis needed)
python3 reports/controller_pulse/make_figs.py
cd reports/controller_pulse && latexmk -pdf pulse_velocity_analysis.tex

make_figs.py writes every number quoted here to numbers.json, and the table rows to design_table.tex.
```